ANATOMY OF AI AGENTS 2026

www.arianprabowo.com · 1. Silicon → 10. Swarms × 4 Organisms

① COLUMNS (1 → 10): 1. Silicon → 2. Python → 3. Data → 4. Pre-Train → 5. SFT → 6. Post-Train → 7. Test-Time → 8. Context → 9. Harness → 10. Swarms
② ROWS (4 ORGANISMS): Core LLM Spine + Separated RAG, Multimodal & Embodied Organisms
③ CELL ANATOMY: Brand-Agnostic Category (H2) + Mechanisms (H3) + ⚙ PRODUCTS & OSS Box (H4) + ⚠ ANTI-PATTERNS Box (H4)

*Synthesized from Haggai Roitman's 'The Hitchhiker's Guide to Agentic AI: From Foundations to Systems' (arXiv:2606.24937, v1.4, June 2026) + 2026 systems tooling · Companion to AI Skill Tree 2026 & AI Ladder Evolution by Arian Prabowo · Updated: 2026-11-08 02:53:22 AEDT (UTC+11:00)

ANATOMY

1 → 10 Stages

Category (H2) · Mechs (H3)
⚙ Products · ⚠ Traps (H4)

1. Silicon

Physical Compute Silicon, GPU Clusters, HBM3e Memory & Interconnect Fabrics

- Unit: FLOPs, HBM Bytes & Warps
- Guide: Ch. 1.6, 2.3, 11.10–14, 30.5

2. Python

Python Runtimes, Package Managers, Virtual Environments & Tensor Compilers

- Unit: Wheels, Tensors & Paged KV
- Guide: Ch. 1.24-25, 11.1-9, 21.5, 22.1

3. Data

Web & Code Corpora, Subword Tokenization & Curation Pipelines

- Unit: Corpora, Subwords & Shards
- Guide: Ch. 1.2, 3.1, 3.3-3.5, 4.1, 30.2

4. Pre-Train

Foundation Model Architectures, Sparse MoE Routing & Self-Supervised Pre-Training

- Unit: Transformer Blocks & MoE
- Guide: Ch. 1.1, 2.2, 3.1, 11.1-9, 30.1

5. SFT

Supervised Fine-Tuning, Instruction Templates, Sequence Packing & LoRA Adapters

- Unit: ChatML, Packing & LoRA
- Guide: Ch. 1.8, 3.2, 7.6, 9.4, 30.2

6. Post-Train

Preference Alignment (RLHF/DPO) & Verifiable-Reward Reinforcement Learning (RLVR/GRPO)

- Unit: Pairs, Rollouts & Advantages
- Guide: Ch. 1.8, 3.2, 12.1-8, 30.2

7. Test-Time

Test-Time Compute Scaling, System-2 Chain-of-Thought Reasoning & Tree Search

- Unit: <think> Tokens, PRM & Trees
- Guide: Ch. 13.1-8, 15.1-9, 30.2

8. Context

Context Engineering, Four-Tier Memory, Tool Calling, MCP & Skills

- Unit: Context, Memory & JSON-RPC
- Guide: Ch. 17, 18.2-5, 22, 23

9. Harness

Single-Agent Harness Engineering, Observe-Decide-Act Loops & Stateful Graph Workflows

- Unit: State, Actions & Git Diffs
- Guide: Ch. 16, 19, 20, 27

10. Swarms

Multi-Agent Orchestration, Hierarchical Swarms, A2A Protocols & Shared State

- Unit: A2A Tasks & Swarm State
- Guide: Ch. 1.14, 14, 24, 26, 28, 30.10

CORE SPINE

LLM

Autoregressive Text, Code & Reasoning Language Models

Token-by-Token Language, Symbolic Math, Code Gen & Autonomous Agent Loops

- 1-3: GPU / PyTorch / BPE
- 4-7: MoE / SFT / RLVR / CoT
- 8-10: MCP / Harness^2 / A2A

Tensor Compute

HBM3e vs. On-Chip SRAM

Tiled Attention & FP8

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- AI ACCELERATORS: H200 · B200 · MI325X
- CLOUD ASICs: TPU v6e · Trainium2
- KERNEL LANGUAGES: CUDA · Triton · CUTLASS
- ATTENTION KERNEL: FlashAttention-3
- CLUSTER FABRIC: NVLink 5 · InfiniBand
- LOW-BIT ENGINE: TransformerEngine FP8

⚠ ANTI-PATTERNS & TRAPS

- × HBM Bandwidth Wall
- × CUDA OOM Crashes
- × Silent SDC Bit-Flips
- × Straggler GPU Stall

Serving Runtime

Deterministic Lockfiles

Paged KV & Graph Capture

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- ENV & PKG MGR: uv · Pixi · Conda · pip
- TENSOR FRAMEWORK: PyTorch 2.x · JAX
- SERVING ENGINE: vLLM · SGLang
- C++ LLM RUNTIME: TRT-LLM · llama.cpp
- GRAPH COMPILER: torch.compile · XLA
- CODE SANDBOX: E2B · Firecracker VM

⚠ ANTI-PATTERNS & TRAPS

- × CUDA ABI & Wheel Hell
- × Unsandboxed Host RCE
- × KV-Cache Preemption
- × pip Breaking flash-attn

Text & BPE Corpus

MinHash Dedup & Filter

Byte-Level BPE Tokenize

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- WEB TEXT CORPORA: FineWeb-Edu · Dolma
- CODE & MATH SETS: The Stack v2 · DCLM
- BPE TOKENIZERS: Tiktoken · SentencePiece
- CURATION ENGINES: Datatrove · NeMo Curat.
- SYNTHETIC CORPORA: Nemotron-CC · Cosmop.
- PII & QUALITY: Presidio · FastText

⚠ ANTI-PATTERNS & TRAPS

- × Synthetic Model Collapse
- × Test-Set Contamination
- × Tokenizer Glitch Tokens
- × PII Baked into Weights

Sparse MoE & Attn

Routed MoE & Latent KV

3D/4D Tensor Parallel

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- OPEN MoE MODELS: DeepSeek-V3 · Llama 4
- DENSE & MoE LLMs: Qwen 3 · Mistral Large
- 3D/4D TRAINERS: Megatron-LM · TorchTitan
- SHARDED DATA PAR: DeepSpeed ZeRO · FSDP2
- SSM / HYBRID ARCH: Mamba-2 · Jamba · RWKV
- COMPACT DENSE: Gemma 3 · Nemotron-4

⚠ ANTI-PATTERNS & TRAPS

- × Loss Spikes & NaN Crash
- × MoE Routing Collapse
- × Pipeline Bubble Stall
- × RoPE Length Collapse

Instruction Tuning

Chat Template & Packing

LoRA / QLoRA & Masking

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- SFT FRAMEWORKS: TRL · Axolotl · Unsloth
- MODULAR TRAINERS: TorchTune · LLaMA-Fact.
- ADAPTER LIBRARIES: HF PEFT · LoRA · DoRA
- WEIGHT MERGING: MergeKit (SLERP/TIES)
- CHAT TEMPLATES: ChatML · ShareGPT · Tulu
- FORMAT BENCHMARK: IFEval · Arena-Hard

⚠ ANTI-PATTERNS & TRAPS

- × Unmasked Prompt Loss
- × EOS/BOS Template Loops
- × Packed Attention Bleed
- × Catastrophic Forgetting

Alignment & RLVR

Group-Relative RL (GRPO)

Pairwise Preference DPO

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- RLVR REASONERS: DeepSeek-R1 · Open-R1
- RLVR FRAMEWORKS: veRL · OpenRLHF · NeMo
- PREFERENCE TUNER: TRL DPO / KTO / SimPO
- REWARD & ASYNC RL: Prime-RL · Skywork-RM
- VERIFIABLE EVALS: LiveCodeBench · AIME
- ALIGNMENT EVALS: RewardBench · AlpacaEv.

⚠ ANTI-PATTERNS & TRAPS

- × Reward Hacking & Bloat
- × Policy Entropy Collapse
- × Zero-Variance Advantage
- × Judge Sycophancy Bias

System-2 Search

Deliberative CoT Tokens

Process RM & Tree Search

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- FRONTIER REASONER: OpenAI o3 / o4-mini
- OPEN REASONERS: DeepSeek-R1 · QwQ-32B
- EXTENDED THINKING: Claude 4.6 · Gemini 2.5
- TREE & BUDGET: LATS · s1 Budget-Force
- PROCESS VERIFIER: Math-Shepherd · SkyPRM
- SYSTEM-2 EVALS: ARC-AGI-2 · AIME · GPQA

⚠ ANTI-PATTERNS & TRAPS

- × 10x Overthinking Burn
- × PRM Verifier Hacking
- × Self-Correction Regress
- × Unfaithful Post-Hoc CoT

Memory & Tool Bus

4-Tier Memory Hierarchy

JSON-RPC Tool & Skill Bus

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- TOOL PROTOCOL: Model Context Protocol
- AGENTIC MEMORY: Letta (MemGPT) · Mem0
- TEMPORAL GRAPH: Zep · LangMem · Cognee
- PROGRESSIVE SKILL: SKILL.md · AGENTS.md
- KV PREFIX CACHE: Anthropic Prompt Cache
- MEMORY BENCHMARK: LoCoMo · LongMemEval

⚠ ANTI-PATTERNS & TRAPS

- × Lost-in-the-Middle Rot
- × 100+ MCP Schema Bloat
- × SKILL.md / MCP Rug-Pull
- × Poisoned Memory State

Observe-Act Loop

Stateful Graph & Retries

Sandboxed Test-Fix Loop

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- CLI CODE AGENTS: Claude Code · Codex
- IDE CODE AGENTS: Cursor · Windsurf · Cline
- STATEFUL GRAPHS: LangGraph · PydanticAI
- OPEN SWE AGENTS: OpenHands · SWE-agent
- DURABLE WORKFLOW: Temporal · Inngest
- CODING BENCHMARK: SWE-bench Verified

⚠ ANTI-PATTERNS & TRAPS

- × Deleting Failing Tests
- × Infinite Loop Token Burn
- × God-Node State Bloat
- × IDE Auto-Approve RCE

Multi-Agent Graph

Orchestrator-Worker DAG

Agent-to-Agent RPC Bus

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- INTER-AGENT RPC: Google A2A Protocol
- AGENT-USER UI BUS: AG-UI · CopilotKit
- ROLE SWARM SDKs: CrewAI · AutoGen (AG2)
- GRAPH SWARM SDKs: LangGraph Swarm · OAI
- SANDBOX WORKTREE: Git Worktree · Dagger
- ROLEPLAY SWARMS: ChatDev · MetaGPT

⚠ ANTI-PATTERNS & TRAPS

- × Swarm Echo Hallucinate
- × $O(N^2)$ Chatty Token Burn
- × Confused Deputy Breach
- × Worktree Merge Conflict

ORGANISM · KNOWLEDGE

RAG

Hybrid Vector, Lexical & Graph Retrieval-Augmented Generation

External Grounding, Dense + BM25 Search, GraphRAG & Citation Verification

- 1-3: SIMD / FAISS / Docling
- 4-7: BGE-M3 / Rerank / HyDE
- 8-10: Qdrant / Agentic RAG

Vector & SSD I/O

GPU Graph & SIMD Scan

Direct NVMe Disk I/O

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- GPU VECTOR INDEX: NVIDIA cuVS · RAFT
- CPU SIMD KERNELS: Intel AVX-512 · AMX
- ASYNC DISK I/O: NVMe Gen5 · io_uring
- ON-DISK GRAPH: DiskANN · SPANN
- DMA STORAGE: GPUDirect Storage
- MEMORY POOLING: RoCEv2 · CXL 3.0

⚠ ANTI-PATTERNS & TRAPS

- × SSD Random-Read Thrash
- × HNSW Exceeding DRAM
- × PCIe Index Bottleneck
- × NUMA Socket Latency

Index Bindings

Zero-Copy Columnar IPC

Async Vector/DB Drivers

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- ANN LIBRARIES: FAISS · HNSWLib · Lance
- COLUMNAR MEMORY: Apache Arrow · Parquet
- IN-PROCESS OLAP: DuckDB · Polars
- ASYNC SQL DRIVER: psycopg3 · asyncpg
- VECTOR CLIENTS: qdrant-client · pymilvus
- LEXICAL & EMBED: Tantivy · BM25S · ONNX

⚠ ANTI-PATTERNS & TRAPS

- × Python GIL Chunk Lock
- × Pickle Index RCE Exploit
- × DB Pool Exhaustion
- × Pandas Copy RAM Blowup

Doc Chunking & ETL

Layout-Aware & AST Split

ACL Metadata & CDC Sync

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- PDF LAYOUT ETL: Docling · Marker PDF
- DOC PARSERS: LlamaParse · Unstruct.
- CODE AST CHUNKER: Tree-sitter AST Split
- ACADEMIC OCR: Grobid · Nougat OCR
- SEMANTIC CHUNKER: Chonkie · LangChain
- CDC STREAM SYNC: Debezium · Airbyte

⚠ ANTI-PATTERNS & TRAPS

- × Mid-Table Naive Split
- × PDF Column Scramble
- × Stale Deleted Vectors
- × Missing ACL Metadata

Dense Bi-Encoders

Contrastive InfoNCE Loss

Late-Interaction & MRL

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- DENSE EMBEDDERS: BGE-M3 · NV-Embed-v2
- LATE INTERACTION: ColBERTv2 · PLAID
- SPARSE LEXICAL: SPLADE-v3 · BM25S
- MATRYOSHKA EMBED: Nomic-Embed · GTE
- COMMERCIAL EMBED: Cohere Embed v3 · E5
- RETRIEVAL EVALS: MTEB · BEIR Benchmark

⚠ ANTI-PATTERNS & TRAPS

- × Embedding Anisotropy
- × False-Negative Batches
- × Domain Acronym Blindness
- × 512-Token Truncation

Retriever Tuning

Hard-Negative Cross-Enc

Distractor-Robust RAFT

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- CROSS-ENCODERS: bge-reranker · Cohere
- GROUNDED SFT: RAFT · RankZephyr
- UNIFIED EMBED+GEN: GritLM · Instructor
- EMBEDDING TUNER: Sentence-Transformers
- LIGHT RERANKERS: FlashRank · Rerankers
- RAG SFT EVALS: RAGBench · RGB Eval

⚠ ANTI-PATTERNS & TRAPS

- × Memorizing Stale Facts
- × Believing Distractors
- × Fake [1] Citation Tags
- × Narrow Query Overfit

Search & Cite RL

Multi-Hop Search RLVR

NLI Faithfulness Reward

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- SEARCH RL MODELS: Search-R1 · R1-Searcher
- QUERY REWRITE RL: DeepRetrieval · ReZero
- NLI VERIFIERS: AlignScore · LettuceDet.
- RAG JUDGE SUITES: RAGAS · TruLens · ARES
- MULTI-HOP QA SET: HotpotQA · MuSiQue
- HALLUCINATION QA: RAGTruth · HaluEval

⚠ ANTI-PATTERNS & TRAPS

- × Redundant Search Loops
- × Cherry-Picking Evidence
- × Over-Abstention Refusal
- × Uncited Extrapolation

Iterative Search

Query Decomp & Rewrite

Reciprocal Rank & Verify

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- REFLECTIVE RAG: Self-RAG · CRAG
- INTERLEAVED CoT: IRCoT · FLARE · HyDE
- NEURAL RERANKER: Cohere Rerank 3.5
- TOKEN RERANKER: ColBERT PLAID Rerank
- LISTWISE LLM RANK: RankGPT · MonoT5
- REASONING RAG QA: BRIGHT · FRAMES Eval

⚠ ANTI-PATTERNS & TRAPS

- × Infinite Rewrite Loops
- × HyDE Hallucination Drift
- × Reranker p99 SLA Blowup
- × Multi-Hop Error Cascade

Hybrid & Graph KB

Dense + Lexical Fusion

Entity-Relation Graph KB

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- VECTOR DATABASES: Qdrant · Milvus · Weav.
- HYBRID SQL / DOC: pgvector · Elasticsearch
- GRAPH RAG ENGINE: MS GraphRAG · LightRAG
- PROPERTY GRAPHS: Neo4j · Memgraph
- PROMPT COMPRESS: LLMLingua-2 · REFRAG
- RAG ORCHESTRATOR: LlamaIndex · Haystack

⚠ ANTI-PATTERNS & TRAPS

- × Indirect Prompt Inject
- × Cross-Tenant Vector Leak
- × Context Stuffing Noise
- × GraphRAG Indexing Cost

Plan-Verify RAG

Adaptive Query Routing

Citation & Span Verify

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- DEEP RESEARCHERS: OpenAI Deep Research
- SEARCH SYNTHESIS: Perplexity Pro · STORM
- AGENTIC RAG DAG: LangGraph CRAG · DSPy
- AGENT SEARCH API: Tavily · Exa · Firecrawl
- FAITHFULNESS GATE: Patronus Lynx · NeMo G.
- RESEARCH BENCH: GAIA · FRAMES Eval

⚠ ANTI-PATTERNS & TRAPS

- × Rephrase Search Thrash
- × Silent Hallucination
- × Citation Laundering
- × Unbounded Research Lag

Federated KB Mesh

Cross-Silo Query Router

OAuth ACL Token Pass

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- ENTERPRISE SEARCH: Glean · Onyx (Danswer)
- MULTI-AGENT RAG: LlamaIndex AgentWorkfl.
- FEDERATED ROUTER: Federated-MCP · GraphQL
- ZANZIBAR RBAC: SpiceDB · OpenFGA ACL
- FEDERATED LAKE: Apache Iceberg · Trino
- ENTERPRISE BENCH: Enterprise RAG-Bench

⚠ ANTI-PATTERNS & TRAPS

- × Circular Citation Loop
- × Cross-Dept ACL Leakage
- × Duplicate Doc Fetching
- × Dropping Minority Facts

ORGANISM · SENSORY

Multimodal

Vision, Audio, Speech, Video & Native Omni-Modal Models

ViT Patch Perception, Native Speech Codecs, Spatio-Temporal Video & Cross-Modal Omni

- 1-3: NVDEC / FFmpeg / Pairs
- 4-7: SigLIP / LLaVA / Visual CoT
- 8-10: Token Merge / LiveKit

Media Codec DSP

HW Video/Audio Decode

3D Patch & Conv Tiling

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- GPU MEDIA CODEC: NVIDIA NVDEC / NVENC
- UNIFIED RAM SoC: Apple M4 Media Engine
- MOBILE / EDGE DSP: Qualcomm Hexagon NPU
- ZERO-COPY VIDEO: GPUDirect Video DMA
- VISION KERNELS: cuDNN · TensorRT-MM
- CAMERA / RTC BUS: MIPI CSI-2 · WebRTC

⚠ ANTI-PATTERNS & TRAPS

- × CPU Video Decode Stall
- × 4K Frame PCIe Flood
- × Audio Sample Clock Drift
- × Video Thermal Throttle

Stream Pipelines

Zero-Copy Tensor Decode

Async Media Ring Buffers

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- VIDEO DEMUXING: FFmpeg · PyAV · Decord
- VISION TRANSFORMS: OpenCV · torchvision
- AUDIO DSP LIBS: torchaudio · librosa
- GPU DATA LOADER: NVIDIA DALI · Kornia
- FAST IMAGE I/O: Pillow-SIMD · Albument.
- SPEECH & VAD: whisper.cpp · Silero

⚠ ANTI-PATTERNS & TRAPS

- × OpenCV BGR vs RGB Bug
- × FFmpeg Zombie Leaks
- × Audio Buffer Underruns
- × Pillow Bomb DoS Crash

Modal Alignment

Contrastive Pair Filter

Scene Cut & Diarization

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- IMAGE-TEXT SETS: DataComp · OBELICS
- DENSE CAPTIONS: ShareGPT4V · LAION-5B
- SPEECH DIARIZE: WhisperX · pyannote
- NEURAL CODECS: EnCodec · Mimi · DAC
- VIDEO SCENE CUT: PySceneDetect · CLIP
- DOCUMENT OCR: PaddleOCR · Surya

⚠ ANTI-PATTERNS & TRAPS

- × Mismatched Alt-Text
- × Low-DPI OCR Misreads
- × Demographic Bias Skew
- × Audio Crosstalk Bleed

Omni Backbones

Early-Fusion Any-to-Any

Flow-Matching & 3D RoPE

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- VISION ENCODERS: SigLIP-2 · DINOv2 ViT
- OPEN VLM BASES: Qwen2.5-VL · InternVL3
- NATIVE OMNI LLMs: GPT-4o · Gemini 2.5 Pro
- FLOW / DiT GEN: Flux.1 · Wan2.1 DiT
- FULL-DUPLEX OMNI: Moshi · Chameleon
- MULTIMODAL EVAL: MMMU · MathVista

⚠ ANTI-PATTERNS & TRAPS

- × Modality Text-IQ Drop
- × Bag-of-Patches Blindness
- × 4K Quadratic ViT Blowup
- × Small-Text Patch Blur

Visual & Voice SFT

Dynamic High-Res Tiling

Bounding-Box & Voice SFT

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- ANYRES VLM SFT: LLaVA-OneVision · Molmo
- SPEECH-LLM SFT: Qwen2-Audio · Ultravox
- COMPACT VLM SFT: PaliGemma 2 · Pixtral
- GROUNDING BBOX: Florence-2 · GroundDINO
- VISUAL PROMPTING: Set-of-Mark (SoM)
- DOC & CHART EVAL: DocVQA · ChartQA

⚠ ANTI-PATTERNS & TRAPS

- × Blind Visual Guessing
- × Projector Detail Loss
- × BBox Coordinate Drift
- × Voice SFT Hurting Math

Perceptual RL

IoU & OCR Verifiable RL

Diffusion & Fact DPO

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- VISUAL RLVR MODELS: R1-OneVision · VLM-R1
- VLM FACT ALIGN: RLAIF-V · POVID · BPO
- DIFFUSION RL: Diffusion-DPO · FlowGRPO
- MULTIMODAL RLHF: InternVL-RL · LLaVA-RL
- OBJECT HALLUC QA: POPE · HallusionBench
- VLM SAFETY EVALS: MM-SafetyBench · SPA-VL

⚠ ANTI-PATTERNS & TRAPS

- × Text-Prior Blind Guess
- × Oversaturated Image Hack
- × Benign Image Refusals
- × Voice Barge-In Jitter

Active Perception

Iterative Crop & Zoom

Best-of-N Visual Verify

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- VISUAL CoT ZOOM: OpenAI o3 Visual CoT
- GUIDED CROP SEARCH: V* (V-Star) Search
- VISUAL PROGRAM: ViperGPT · VisProg
- TOOL-USING VLMs: LLaVA-Plus · CogCom
- DIFFUSION SCALING: ControlNet Tile · CFG
- PERCEPTION EVALS: BLINK · V-Bench Eval

⚠ ANTI-PATTERNS & TRAPS

- × Zooming Irrelevant Bg
- × Blurry Crop Hallucinate
- × High-Res Token Overflow
- × CFG Over-Step Artifacts

Token & KV Pruning

ViT Patch Merge & Drop

Streaming Video KV Ring

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- PATCH MERGING: ToMe (Token Merging)
- VLM TOKEN PRUNE: FastV · PruMerge
- STREAMING KV RING: StreamingLLM · InfiniPot
- VIDEO INDEX API: TwelveLabs · Marengo
- AUDIO-TEXT ALIGN: Whisper Timestamp Idx
- LONG VIDEO EVALS: Video-MME · MLVU Eval

⚠ ANTI-PATTERNS & TRAPS

- × 10m Video Filling 128k
- × Pruning Tiny OCR Text
- × Base64 JSON-RPC Crash
- × Audio-Video Sync Drift

Duplex Media Loop

Full-Duplex VAD Barge-In

Render-Inspect-Fix Loop

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- VOICE HARNESSES: LiveKit Agents · Pipecat
- DUPLEX SPEECH API: OpenAI Realtime API
- NODE GEN GRAPH: ComfyUI · InvokeAI
- LOW-LATENCY TTS: ElevenLabs · Cartesia
- MEDIA PROVENANCE: C2PA · SynthID
- DUPLEX BENCHMARK: Full-Duplex-Bench

⚠ ANTI-PATTERNS & TRAPS

- × Talking Over User Voice
- × Transcribing Own TTS
- × VLM Breaking CSS Layout
- × >3s Cascaded Voice Lag

Cross-Modal Studio

Director-Storyboard DAG

Multi-Track Sync & QA

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- VIDEO GEN MODELS: Runway Gen-4 · Sora
- CINEMA DI-T MODELS: Google Veo 2 · Kling
- AUDIO / MUSIC GEN: Suno · Udio · StableAud.
- WORKFLOW CLOUD: ComfyUI Swarm · Fal.ai
- TIMELINE ASSEMBLY: FFmpeg · OTIO Timeline
- VIDEO GEN BENCH: MovieGenBench · VBench

⚠ ANTI-PATTERNS & TRAPS

- × Character Identity Drift
- × Voice Promising Failed Op
- × Audio-Video Lip Desync
- × Runaway Render Spend

ORGANISM · ACTION

Embodied

GUI Computer Use, Physics World Models & VLA Robotics

Browser/OS GUI Control, 3D Physics Simulators, World Models & VLA Robot Policies

- 1-3: Edge NPU / Isaac / Teleop
- 4-7: VLA π0 / ACT / Sim2Real
- 8-10: 3DGS / OSWorld / Fleets

Edge RT Compute

1kHz Sensor ISP & Bus

GPU Rigid/Soft Physics

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- ROBOTICS SoC: NVIDIA Jetson Thor
- GPU PHYSICS SIM: PhysX 5 · NVIDIA Warp
- REAL-TIME FIELD: EtherCAT · CAN-FD
- LOW-POWER NPU: Qualcomm RB5 · Hailo
- 3D LIDAR SENSOR: Ouster · Velodyne
- DEPTH CAMERAS: RealSense · ZED 2i

⚠ ANTI-PATTERNS & TRAPS

- × Edge Thermal Throttle
- × OS Jitter Breaking 1kHz
- × CSI Camera Frame Drop
- × Sim Contact Mismatch

Middleware & IPC

Pub-Sub DDS & Zero-Copy

Browser & Physics APIs

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- ROBOT MIDDLEWARE: ROS 2 · CycloneDDS
- BROWSER CONTROL: Playwright · Chrome CDP
- RL ENV INTERFACE: Gymnasium · PettingZoo
- PHYSICS ENGINES: MuJoCo · Isaac Lab
- KINEMATICS & 3D: Pinocchio · Open3D
- ROBOT SDK & BUS: LeRobot · Eclipse Zenoh

⚠ ANTI-PATTERNS & TRAPS

- × ROS 2 Multicast Storms
- × Zombie Browser Leaks
- × Euler Gimbal Lock Bug
- × Sim Step Drift Desync

Trajectory Logs

Teleop & Action Sync

DOM + Screen Traces

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- ROBOT TRAJECTORY: Open X-Embodiment
- MANIPULATION SET: DROID · BridgeData V2
- EGOCENTRIC VIDEO: Ego4D · Epic-Kitchens
- GUI ACTION LOGS: Mind2Web · OSWorld
- BIMANUAL TELEOP: ALOHA · GELLO Rig
- TRAJECTORY STORE: RLDS · LeRobot Parquet

⚠ ANTI-PATTERNS & TRAPS

- × Teleop Hesitation Noise
- × Screen Resolution Drift
- × Action-Space Mismatch
- × Unlogged Confounders

World & VLA Base

Latent Physics Prediction

Visuomotor Token Fusion

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- FLOW VLA MODELS: π0 (Pi-Zero) · OpenVLA
- GENERALIST VLA: RT-2 · Octo · CrossFormer
- WORLD MODELS: V-JEPA 2 · NVIDIA Cosmos
- INTERACTIVE SIM: Google Genie 2 · UniSim
- 3D POINT ENCODER: PointNet++ · Uni3D
- EMBODIED SUITES: Habitat 3.0 · ManiSkill

⚠ ANTI-PATTERNS & TRAPS

- × Spurious Causal Confusion
- × World Physics Delusion
- × Mode-Averaging Freeze
- × Camera Viewpoint Fragile

Behavior Cloning

k-Step Action Chunking

GUI Pixel-to-Action SFT

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- ACTION CHUNKING: ACT · ALOHA Policy
- DIFFUSION POLICY: Diffusion Policy · DP3
- GUI GROUNDING: OS-Atlas · SeeClick
- GUI ACTION MODELS: ShowUI · Aguvis · CogAg.
- BIMANUAL VLA SFT: RDT-1B · Octo Tune
- BC BENCHMARKS: LIBERO · Meta-World

⚠ ANTI-PATTERNS & TRAPS

- × Compounding BC Drift
- × Copycat Action Inertia
- × Lighting Overfitting
- × Static Coordinate Click

Sim-to-Real RL

Domain-Randomized PPO

Residual & Safe HIL-RL

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- GPU RL TRAINERS: Isaac Gym · CleanRL
- REAL-WORLD HIL RL: HIL-SERL · RLPD
- LLM REWARD SYNTH: Eureka · DrEureka
- WEB / OS RL ENVS: WebArena · OSWorld-RL
- LEGGED LOCOMOTION: RSL-RL · Legged-Gym
- SAFE CONTROL RL: Safe-RL · ControlBarrier

⚠ ANTI-PATTERNS & TRAPS

- × Simulator Physics Exploit
- × Hardware Damage in RL
- × Reward-Hacking 'Buy' Btn
- × Sparse Credit Stall

Predictive Rollout

Receding-Horizon MPC

World-Model Tree Search

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- SAMPLING PLANNER: MPPI · CEM Rollouts
- LATENT WORLD MPC: TD-MPC2 · DreamerV3
- 3D VALUE MAPS: VoxPoser · Code-as-Pol.
- AFFORDANCE PLAN: SayCan · InnerMonologue
- MOTION PLANNERS: MoveIt 2 · Drake MPC
- GUI TREE SEARCH: AgentQ · WebTree MCTS

⚠ ANTI-PATTERNS & TRAPS

- × Planning Lag > 1kHz Loop
- × 5s Rollout Physics Drift
- × Unverified Destructive Click
- × Obstacle Deadlock Loop

Spatial Scene Map

3D Voxel & Gaussian Map

Pruned DOM & AXTree

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- RADIANCE FIELDS: 3D Gaussian Splatting
- VOXEL OCCUPANCY: OctoMap · nvblox
- 3D SCENE GRAPHS: Hydra · ConceptGraphs
- DOM DISTILLATION: Browser-Use DOM Pruner
- ACCESSIBILITY TREE: Playwright AXTree · SoM
- SLAM LOCALIZATION: RTAB-Map · ORB-SLAM3

⚠ ANTI-PATTERNS & TRAPS

- × 50k-Node Raw DOM Flood
- × Dynamic Obstacle Ghosts
- × Out-of-Sight Forgetting
- × Odometry Frame Drift

Closed-Loop Act

Screen-Action-Verify

2Hz VLM + 50Hz Control

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- DESKTOP AGENTS: Anthropic Computer Use
- BROWSER AGENTS: OpenAI Operator · CUA
- GUI PARSER & VLA: UI-TARS · OmniParser v2
- WEB AUTOMATION: Browser-Use · Skyvern
- ROBOT NAV STACK: ROS 2 Nav2 · MoveIt 2
- OS AGENT BENCH: OSWorld · WindowsArena

⚠ ANTI-PATTERNS & TRAPS

- × Fake Pop-Up Injection
- × Infinite 'Pay' Clicks
- × UI Lag Misaligned Click
- × Ignoring Gripper Slip

Fleet Coordination

Conflict-Free Path Plan

Shared Spatial Twin

⚙ SUB-CATEGORY ▸ PRODUCTS & OSS

- FLEET MIDDLEWARE: Open-RMF Robotics
- DIGITAL TWIN CTRL: NVIDIA Isaac Mission
- MAPF PATH SOLVER: CBS / LaCAM Solvers
- FLEET TELEMETRY: Foxglove · Formant
- INTEROP STANDARD: VDA 5050 · AWS RoboRun.
- SWARM SIM BENCH: Flatland · VMAS Eval

⚠ ANTI-PATTERNS & TRAPS

- × Corridor Traffic Gridlock
- × Faulty Sensor Map Poison
- × Wi-Fi Drop Fleet Split
- × Dual-Arm Grasp Collision